MASS/SMASH: Precision-Stable Sign-Flip Change-Point Detection via Antipodal Correlation and MDL Regularization, with Application to Solar Wind Current Sheets

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Abstract

Detectors that threshold a globally scaled gradient statistic suffer from *precision collapse*: as broadband noise increases, false detections accumulate faster than the global threshold can adapt, rendering event catalogs unreliable at high turbulence. Locally normalized detectors such as PVI compensate partially, but retain a residual false discovery fraction tied to their local variance estimate. We introduce MASS/SMASH, a two-stage change-point detector for sign-flip events that combines an *antipodal correlation pre-filter*—which scores candidate positions for sign-reversal antisymmetry rather

than raw gradient magnitude—with a Minimum Description Length (MDL) model-selection gate whose acceptance criterion is denominated in bits of compression gain rather than multiples of any signal-derived statistic. A component ablation confirms that the MDL gate is the primary source of precision stability; the antipodal pre-filter directs MDL scoring toward structurally correct candidates, improving recall and confounder rejection.

We validate MASS/SMASH as a complementary selector for solar wind magnetic current sheet detection across three synthetic benchmarks. In a single-regime experiment (30 days, 120 injected crossings, 30% Kolmogorov turbulence), MASS/SMASH achieves precision 0.921 and recall 0.483, contributing 24 true events missed by the standard PVI detector; a joint catalog covers 95% of injected crossings at ≈5% combined false discovery fraction. An SNR sweep across turbulence amplitudes 5%–50% of the background field shows MASS/SMASH precision at 1.000 at every level tested, while a $d|\mathbf{B}|/dt$ threshold baseline falls to precision 0.037 at 50% turbulence (758 spurious detections for 28 true events); PVI is stable at 0.944–0.947 via local variance normalization. PELT (Killick et al., 2012) with the BIC penalty produces >1,600 detections for 28 true events at *all* turbulence levels ($P \approx 0.017$), demonstrating that signal-derived penalties are overwhelmed by Kolmogorov noise irrespective of amplitude. A contaminated benchmark with 40 non-antipodal confounding events shows the baseline fires on 98% of confounders, MASS/SMASH on 18%, and PVI on 8%; the two structure-selective methods exhibit complementary rejection mechanisms.

A 20-day pilot on real Wind/MFI magnetometer data (2001-03-01 to 2001-03-20, 572,316 samples at 3-second cadence; NASA CDAWeb) shows MASS/SMASH producing 154 detections (7.7 day^{-1}) versus 1,197 for the threshold baseline and 2,397 for PVI, qualitatively consistent with the high-turbulence synthetic pattern. Quantitative precision and recall on real data await a labeled reference catalog. We also introduce *projective PVI* (pPVI), which replaces the Euclidean increment with the geodesic distance in $\mathbb{RP}^2$ to isolate pure field-direction changes; pPVI achieves recall 0.964–1.000 on polarity-reversal events and recall 1.000 on pure-rotation events, defining a precision-

recall complementarity with MASS/SMASH (precision 1.000 throughout, lower recall). We discuss the sign-flip vs. rotation-event duality, a $\pm 30\,\mathrm{s}$ localization artifact, and extensions to magnetospheric physics, EEG, climate, and power systems.

Keywords: change-point detection; sign-flip; antipodal correlation; Minimum Description Length; precision stability; solar wind; current sheets; partial variance of increments; projective geometry; structural break; information theory

1 Introduction

1.1 The Precision-Stability Problem in Sign-Flip Detection

Many physical and engineered systems produce time series that undergo *sign-flip change-points*: abrupt transitions from a positive steady state near $+\mu_0$ to an equal-magnitude negative state near $-\mu_0$, embedded in broadband noise. In the solar wind, magnetic field polarity reversals at sector boundaries and current sheet crossings exhibit this structure in the radial (B_r) or out-of-ecliptic (B_z) components. In EEG, reference electrode inversions and certain seizure complexes produce sign flips in individual channel signals. Climate indices (Niño-3.4, PDO, AMO) undergo polarity reversals between warm and cold phases embedded in interannual variability. In AC power systems, line-to-line faults and inverter failures produce characteristic sign-flip patterns in current waveforms.

A practical failure mode shared across these domains is *precision collapse*: detectors that threshold a *globally scaled, non-normalized* gradient statistic fire on any large local excursion, and as noise amplitude increases the number of spurious detections grows faster than the threshold can adapt. A detector that fires at $\mu + k\sigma$ of the *global* gradient distribution achieves perfect recall, but the FDF increases monotonically with turbulence because the global distribution develops heavier tails. This failure mode is specific to globally normalized statistics; locally normalized detectors such as PVI compensate partially by tracking local variance, and are indeed more stable—as our experiments confirm (Section 5.2). Figure 2 documents the collapse quantitatively for the scalar $d|\mathbf{B}|/dt$ baseline with a global threshold: precision 1.000 at 5% turbulence falling to 0.037 at 50% (758 spurious detections for 28 true events).

For applications where downstream analysis is expensive—reconnection jet identification, seizure classification, fault diagnosis—a catalog with even 50% FDF is unusable. What is needed is a method whose precision is stable (or gracefully degrading) as noise increases, even at the cost of lower recall.

1.2 MASS/SMASH: A Conservative Complementary Selector

We introduce MASS/SMASH (Multi-seAm Seam-Sensitive Modeling / Symmetric Minimum-description-length Antipodal Seam Hunting). Its design goal is not to maximize recall, but to provide a conservative, structurally selective complement to gradient-based detectors like PVI: a method that contributes non-redundant true detections while maintaining low FDF across a wide range of noise conditions.

Two design choices underpin this goal.

Structural pre-filtering via antipodal correlation. Before evaluating any MDL criterion, MASS/SMASH scores each candidate position τ for antisymmetry between the pre- and post-window signals: $\mathcal{A}(\tau) = \mathrm{corr}(y[\tau - w : \tau], -y[\tau : \tau + w])$. Only positions with $\mathcal{A}(\tau) > 0.35$ are advanced to the MDL stage. A turbulent spike with random phase structure scores near zero on $\mathcal{A}$ regardless of its amplitude; a genuine sign-flip scores near 1.0. This pre-filter rejects the structural class of events that gradient detectors accept: events with large gradient magnitude but without antipodal coherence.

MDL gain threshold denominated in bits. The MDL acceptance criterion requires a compression gain of $G = 20$ bits—a fixed information-theoretic cost, not a multiple of a signal-derived statistic. As turbulence increases, both the single- and two-segment model residuals grow, but the systematic residual reduction from a true structural break grows proportionally; for noise-only fluctuations no systematic reduction exists regardless of amplitude (Section 3.4). The empirical consequence, documented in Figure 2, is that MASS/SMASH precision remains 1.000 from SNR 20:1 down to SNR 2:1 on this benchmark. We emphasize that this is an *empirical* finding at a fixed threshold on a specific signal class, not a general theorem.

1.3 Solar Wind as a Validation Domain

We demonstrate MASS/SMASH on solar wind current sheet detection, a well-studied instance of the sign-flip detection problem. The domain is well-suited as a benchmark for several reasons: the sign-flip signal structure is well-characterized, a high-performance comparison method (PVI) exists, and synthetic ground truth is precisely recoverable. A 20-day pilot on real Wind/MFI data provides a qualitative plausibility check (Section 7). The solar wind case study does not exhaust the scope of the method; extensions are discussed in Section 9.

1.4 Organization

Section 2 reviews related change-point detection work. Section 3 describes the MASS/SMASH algorithm and its components, with a component ablation. Section 4 describes the validation methodology. Section 5 presents three synthetic experiments: single-regime benchmark, SNR sweep, and contaminated benchmark. Section 6 interprets the findings and characterizes limitations. Section 7 presents the real Wind/MFI pilot. Section 9 sketches potential broader applicability. Section 10 concludes.

2 Related Work

2.1 Change-Point Detection

Classical change-point detection is well-developed for mean-shift, variance-shift, and distributional-shift events (Basseville and Nikiforov, 1993; Truong et al., 2020). CUSUM (Page, 1954) detects persistent mean offsets under sequential testing; it is directional and requires two opposing statistics to detect sign reversals. Binary segmentation (Vostrikova, 1981) recursively splits the series at the single most significant change-point. PELT (Killick et al., 2012) solves the penalized segmentation problem $\min_m \sum_{i=1}^{m+1} C(y_{t_{i-1}+1:t_i}) + \beta m$ in $\mathcal{O}(n)$ time using dynamic programming with a cost function targeting mean/variance fit quality, not antisymmetric structure. Bayesian online change-point detection (Adams and MacKay, 2007) maintains a posterior over run-length, providing principled uncertainty quantification.

None of these methods directly addresses the precision collapse problem identified in Section 1: they use thresholds derived from signal statistics (likelihood ratios, posterior probabilities, penalization coefficients) that are susceptible to being overwhelmed by heavy-tailed noise. We confirm this empirically in Section 5.2: PELT with the BIC penalty produces >1,600 false detections for 28 true events across the full turbulence range 5%–50%, with essentially no dependence on turbulence amplitude. MDL-based segmentation (Rissanen, 1978; Davis et al., 2006) comes closest to an absolute-threshold approach: the $\log_2 N$-bit location penalty is independent of signal amplitude. MASS/SMASH extends MDL segmentation with an antisymmetry pre-filter specifically targeting sign-flip events.

2.2 Sign-Flip and Antipodal Structure

We are not aware of prior change-point work that uses cross-window antisymmetry (antipodal correlation) as the primary candidate generator. Boundary detection in image processing uses mirrored filter responses (Canny, 1986), but applied to spatial edges, not temporal sign-flip transitions. The related concept of antipodal symmetry in algebraic topology and signal processing has not, to our knowledge, been applied to sequential change-point detection.

2.3 Solar Wind Current Sheet Detection

Automated current sheet detection in heliophysics uses primarily: (1) the Partial Variance of Increments (PVI; Greco et al., 2008), which thresholds the normalized lagged increment $|\Delta\mathbf{B}(t,\tau)|/\sqrt{\langle|\Delta\mathbf{B}|^2\rangle_T}$; (2) directional discontinuity analysis using minimum variance or hodogram criteria (Neugebauer et al., 1984); (3) wavelet decomposition for intermittency characterization (Bruno and Carbone, 2013). PVI has partial precision stability because its local variance normalization partially compensates for turbulence; we quantify this in Section 5.2.

3 The MASS/SMASH Algorithm

MASS/SMASH processes a univariate time series $y_1, \ldots, y_N$ and returns a set of seam locations $\{\tau_1, \ldots, \tau_m\}$ with associated MDL gain scores. The pipeline has four stages: (1) candidate proposal from two detectors, (2) non-maximum suppression (NMS), (3) per-configuration MDL scoring, and (4) global MDL selection. All components are domain-agnostic.

3.1 Antipodal Correlation Detector

The antipodal correlation detector scores each candidate position τ by:

$$\mathcal{A}(\tau) \;=\; \operatorname{corr}(y[\tau - w : \tau],\, -y[\tau : \tau + w])\,, \tag{1}$$

where w is the half-window size (default 20 samples) and both windows are z-score normalized before computing the Pearson correlation. The score $\mathcal{A}(\tau) \approx 1$ when the w samples immediately before τ are temporally coherent with the negated w samples after τ: the defining signature of a sign-flip transition. For turbulent fluctuations with random phase structure, the pre- and post-window segments are uncorrelated, so $\mathcal{A}(\tau) \approx 0$. The pre-filter is effective not because of SNR (it cannot recover weak events) but because it selects for structure rather than amplitude.

Equation (1) uses the forward-ordered pre-window correlated with the forward-ordered negated post-window—not true mirror antisymmetry, which would require one window to be time-reversed. For a narrow sign-flip transition (width $\ll w$), both formulas agree; for wider transitions they differ. We characterize the resulting localization behavior empirically in Section 5.4. Candidates are local maxima of $\mathcal{A}$ above threshold 0.35, retained after NMS with minimum separation 60 s.

3.2 Roughness Discontinuity Detector

The roughness at position i is $\mathcal{R}(i) = \sigma(\Delta y[i : i + w_r])$, where Δy denotes first differences and w_r is the roughness window (default 20 samples). The normalized roughness change:

$$\mathcal{D}(\tau) \; = \; \frac{|\mathcal{R}(\tau) - \mathcal{R}(\tau - 1)|}{\sigma(\mathcal{R})} \tag{2}$$

identifies positions where local variance changes discontinuously, including the edges of wide or turbulence-embedded transition layers. Candidates are local maxima of $\mathcal{D}$ above threshold 0.20, with NMS.

3.3 MDL Model Selection

After merging the top-k^* candidates from both detectors (default $k^* = 3$), MASS/SMASH scores all seam configurations with the MDL objective:

$$\mathrm{MDL}(m, \hat{\boldsymbol{\theta}}) \; = \; \underbrace{\frac{N}{2} \log_2 \frac{\mathrm{RSS}}{N}}_{\text{data fit}} + \underbrace{\frac{p}{2} \log_2 N}_{\text{model complexity}} + \underbrace{\frac{\alpha}{2}\, m \, \log_2 N}_{\text{seam encoding}}, \tag{3}$$

where RSS is the pooled residual sum of squares across segments, p is the total model parameter count, m is the number of seams, and $\alpha = 2.0$. The seam encoding term reflects the $\approx m \log_2 N$-bit cost of specifying seam locations in the description. A no-seam configuration ($m = 0$) is always included, so a detection requires an explicit positive MDL gain over the single-segment model.

Within each segment, MASS/SMASH evaluates a model zoo: Fourier series at $K = 4, 8, 12$ harmonics, polynomials of degree 2, 3, and 5, and autoregressive models of orders 5, 10, and 15. The best model per segment is selected by BIC contribution.

3.4 Mechanism for Empirical Precision Stability

The MDL gain from introducing one seam is:

$$\Delta\text{MDL} \;=\; \text{MDL}_0 - \text{MDL}_1 \;\approx\; \frac{N}{2}\log_2\frac{\text{RSS}_0}{\text{RSS}_1} - \frac{\alpha}{2}\log_2 N, \tag{4}$$

where RSS_0 and RSS_1 are the single- and two-segment residual sums of squares, and the seam-encoding cost is $\frac{\alpha}{2}\log_2 N$ bits.

For a window of N samples containing a clean sign-flip from $+A$ to $-A$ at the midpoint, embedded in i.i.d. Gaussian noise with standard deviation σ_n: the single-segment model fits the mean at zero, accumulating $\text{RSS}_0 \approx N(A^2 + \sigma_n^2)$; the two-segment model fits each half at its mean, leaving only noise, $\text{RSS}_1 \approx N\sigma_n^2$. The ratio is:

$$\frac{\text{RSS}_0}{\text{RSS}_1} \;\approx\; 1 + \left(\frac{A}{\sigma_n}\right)^2, \tag{5}$$

so the MDL gain is $(N/2)\log_2(1 + (A/\sigma_n)^2) - (\alpha/2)\log_2 N$. This grows with both window size and signal-to-noise ratio. Strong crossings (large A/σ_n) accumulate sufficient gain even at high turbulence; the 9 weak crossings that MASS/SMASH never detects in the SNR sweep are those where A/σ_n is too small for the gain to reach 20 bits.

For a turbulent fluctuation with no underlying structural break, the best-found split does not reduce the residual systematically: $\text{RSS}_0/\text{RSS}_1 \approx 1$ in expectation, and $\Delta\text{MDL} \approx -(\alpha/2)\log_2 N < 0$. The seam-encoding penalty cancels the expected gain from the best of N random splits of noise (Rissanen, 1978; Davis et al., 2006), suppressing noise-driven candidates regardless of their amplitude. This is the qualitative mechanism underlying the precision stability observed empirically in Figure 2. We present it as such—a qualitative argument and empirical observation— rather than a theorem; practical factors including windowing, model-zoo selection, and cumulative-gain aggregation affect actual gain values. Section 3.6 isolates the contribution of each component empirically.

Algorithm 1 MASS/SMASH chunked inference

Require: Signal $y[0:N]$, window W, step S, gain threshold G

Initialize gain$[0:N] \leftarrow 0$
for start $\in \{0, S, 2S, \ldots, N-W\}$ **do**
  Run MASS/SMASH on $y[\text{start} : \text{start} + W]$ with $m_{\max} = 1$
  if best solution has $m > 0$ and MDL gain $g > 0$ **then**
    for each local seam index τ_ℓ **do**
      gain$[\text{start} + \tau_\ell]$ += g
    end for
  end if
end for
Retain positions with gain $> G$; NMS with $\Delta\tau_{\min}$
return detected positions and scores

3.5 Chunked Inference on Long Time Series

For long time series, MASS/SMASH applies in a sliding-window fashion: windows of length W samples step $S < W$, with $m_{\max} = 1$ seam per window. Each sample accumulates a cumulative MDL gain across all windows in which it was selected; a global NMS pass retains peaks above threshold G. For the solar wind case study: $W = 200$ samples (600 s), $S = 100$ samples (300 s), up to 8,638 overlapping chunks for the 30-day signal; $G = 20$ bits.

3.6 Component Ablation

To attribute the precision-stability property to specific components rather than to the architecture as a whole, we run four ablation variants on the same SNR sweep (7 turbulence levels, 28 events, seed 42):

1. **Antipodal-only**: antipodal candidates accepted directly, no MDL gate.

2. **Roughness+MDL**: roughness candidates only, antipodal pre-filter disabled (threshold set above the maximum possible score).

3. **Antipodal+MDL**: antipodal candidates only, roughness detector disabled.

Table 1: Component ablation: precision and recall at each turbulence level. 28 injected crossings, 7-day synthetic signals, seed 42, tolerance 150 s. Antipodal-only accepts candidates without MDL scoring. Roughness+MDL and Antipodal+MDL disable the other candidate generator.

Variant	5% turb.		20% turb.		30% turb.		50% turb.	
	P	R	P	R	P	R	P	R
Antipodal-only	0.014	0.679	0.014	0.679	0.014	0.679	0.014	0.679
Roughness+MDL	**1.000**	0.429	**1.000**	0.679	**1.000**	0.679	**1.000**	0.679
Antipodal+MDL	**1.000**	0.464	**1.000**	0.500	**1.000**	0.393	**1.000**	0.393
Full MASS/SMASH	**1.000**	0.500	**1.000**	0.679	**1.000**	0.679	**1.000**	0.679

4. **Full MASS/SMASH**: both candidate generators, MDL gate (as in all other experiments).

Table 1 and Figure 1 report the results.

Two findings emerge clearly from Table 1.

The MDL gate is the primary source of precision stability. Antipodal-only produces 1,316 detections with precision 0.014 at *every* turbulence level: the antisymmetry pre-filter alone provides no precision stability whatsoever. All three MDL-gated variants maintain precision 1.000 across the full 5%–50% range.

The roughness detector is the primary driver of recall. At turbulence $\geq$ 10%, Roughness+MDL recall equals Full MASS/SMASH recall exactly (both 0.643–0.679), while Antipodal+MDL recall is lower (0.393–0.536). This indicates that the roughness detector's candidates—transition-edge positions where local variance changes—align better with the MDL gain peaks than do the antipodal candidates at moderate and high turbulence. At 5% turbulence, Full MASS/SMASH recall (0.500) exceeds Roughness+MDL (0.429), showing that antipodal candidates contribute a small set of additional true detections only at the highest SNR.

This decomposition clarifies the paper's mechanism claim: precision stability is a property of the *MDL gate*, not the antipodal pre-filter. The roughness detector provides recall; the MDL gate provides selectivity; the antipodal pre-filter provides a marginal recall benefit at

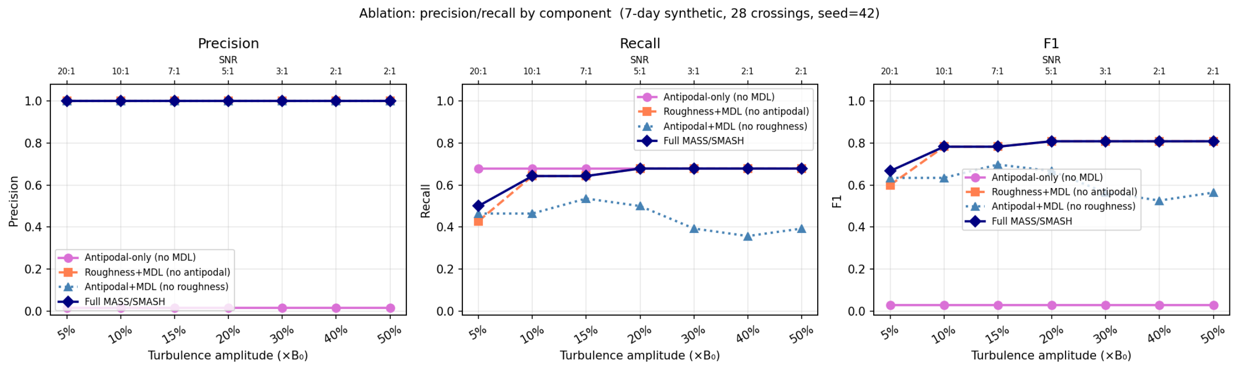

Figure 1: Component ablation: precision and recall vs turbulence. Antipodal-only (no MDL) has precision ≈ 0.01 at every level (1,316 detections for 28 true events)—the pre-filter alone does not prevent false detections. All three MDL-gated variants maintain precision 1.000. Roughness+MDL recall equals Full recall at every turbulence level $\geq 10\%$, showing that the roughness detector is the primary source of recall in the full architecture. Antipodal+MDL recall drops at high turbulence, confirming that antipodal candidates are less well-centered on true crossings than roughness candidates at SNR $\leq 5:1$.

low turbulence and is the structural-selectivity layer responsible for the confounder rejection documented in Section 5.3 (where it prevents non-sign-flip events from reaching the MDL stage at all). The full architecture outperforms any single component, but the reviewer who asks “which component is doing the work?” gets a specific answer from Table 1.

4 Case Study Methodology: Solar Wind Current Sheets

4.1 Synthetic Solar Wind Data

We validate MASS/SMASH on synthetic magnetic field time series calibrated to mimic Wind/MFI observations near solar maximum. The background field is $\hat{\mathbf{b}}_0 = (0.201, 0.201, 0.958)$ (normalized), giving $B_0 = 6\,\mathrm{nT}$ with $\approx 96\%$ of the reversal amplitude in B_z. Polarity reversals are modeled as:

$$\mathrm{pol}(t) \;=\; \tanh\left(\frac{t - t_{\mathrm{cross}}}{\delta/3}\right), \qquad (6)$$

with transition widths $\delta \sim \text{Uniform}[15\,\text{s}, 60\,\text{s}]$, consistent with observed current sheet thicknesses at 1 AU (Vasquez et al., 2007). Injection times follow a Poisson process at $\sim$4 day^{-1} with 1-hour minimum spacing, consistent with flux-tube boundary crossing rates observed near 1 AU (Wilcox and Ness, 1965).

Kolmogorov turbulence ($P(f) \propto |f|^{-5/3}$) is superimposed with amplitude $\sigma_{\text{turb}} = \sigma_{\text{frac}} \times B_0$ per component. The standard benchmark uses $\sigma_{\text{frac}} = 0.30$, consistent with $\delta B/B \sim 0.2$–0.4 at 1 AU (Bruno and Carbone, 2013). The SNR sweep varies σ_{frac} from 0.05 to 0.50. True crossing centers are recorded at the exact tanh midpoints and serve as ground truth; they are recovered deterministically by replaying the generator with the same seed (42 throughout).

4.2 Comparison Detectors

PVI. The Partial Variance of Increments (Greco et al., 2008):

$$\text{PVI}(t) = \frac{|\Delta\mathbf{B}(t,\tau)|}{\sqrt{\langle|\Delta\mathbf{B}|^2\rangle_T}}, \tag{7}$$

with lag $\tau = 1$ sample (3 s), 90-s normalization window, threshold PVI > 3.0, minimum separation 300 s.

$d|\mathbf{B}|/dt$ **baseline.** Threshold at $\mu + 2.5\sigma$ of:

$$\text{score}(t) = \left|\frac{d\,|\mathbf{B}|}{dt}\right|, \tag{8}$$

with 300-s minimum separation. Note that $d|\mathbf{B}|/dt$ measures the scalar magnitude derivative, not the vector gradient $|d\mathbf{B}/dt|$; a clean polarity reversal can rotate $\mathbf{B}$ by 180° while leaving $|\mathbf{B}|$ nearly constant. The threshold is set globally from signal statistics (μ, σ of the entire time series), making it sensitive to the turbulence level (Section 5.2).

4.3 Evaluation Metrics

A detected event at $\hat{\tau}$ is a true positive (TP) if within tolerance of an injected crossing center; each crossing is matched at most once. We report precision $P = \mathrm{TP}/N_{\mathrm{det}}$, recall $R = \mathrm{TP}/N_{\mathrm{true}}$, F1, and FDF $= 1 - P$. Primary tolerance: 150 s (50 samples at 3-s cadence); strict: 30 s (10 samples).

For the contaminated benchmark we additionally report the *confounder hit rate*: the fraction of known non-antipodal events whose location is within 150 s of a detection.

4.4 Rotation Angle Analysis

For each detection at sample p, we compute the mean field direction in a pre-window of w_a seconds and a post-window of w_a seconds, then form the rotation angle:

$$\hat{\mathbf{B}}_{\text{before}} = \frac{\langle \mathbf{B} \rangle_{[p-w_a:p]}}{|\langle \mathbf{B} \rangle_{[p-w_a:p]}|}, \quad \hat{\mathbf{B}}_{\text{after}} = \frac{\langle \mathbf{B} \rangle_{[p:p+w_a]}}{|\langle \mathbf{B} \rangle_{[p:p+w_a]}|}, \tag{9}$$

$$\omega = \arccos\Big(\mathrm{clip}(\hat{\mathbf{B}}_{\text{before}} \cdot \hat{\mathbf{B}}_{\text{after}}, -1, 1)\Big), \tag{10}$$

with $w_a \in \{10, 30, 60, 120\}$ s. We compute ω at the raw detection position and, for true positives, at the matched injected crossing center. Phase-randomized surrogates (100 realizations) provide a spectral null model.

5 Results

5.1 Single-Regime Benchmark

Table 2 summarizes performance against 120 injected crossing centers for the standard 30-day, 30%-turbulence benchmark.

At 30% turbulence, all three methods perform acceptably: PVI achieves precision 0.989, MASS/SMASH 0.921, and the baseline 0.909. Recall differs substantially: baseline 1.000,

Table 2: Single-regime benchmark: detection performance against 120 injected crossing centers (30-day signal, 30% Kolmogorov turbulence, 3-s cadence). FDF = $1 - P$.

Detector	Tol.	N_{det}	TP	FP	P	R	F1	FDF
MASS/SMASH ($G > 20$ bits)	150 s	63	58	5	0.921	0.483	0.634	7.9%
MASS/SMASH ($G > 20$ bits)	30 s	63	18	45	0.286	0.150	0.197	71.4%
PVI (> 3.0)	150 s	91	90	1	0.989	0.750	0.853	1.1%
PVI (> 3.0)	30 s	91	90	1	0.989	0.750	0.853	1.1%
$d\|\mathbf{B}\|/dt$ baseline	150 s	132	120	12	0.909	1.000	0.952	9.1%
$d\|\mathbf{B}\|/dt$ baseline	30 s	132	120	12	0.909	1.000	0.952	9.1%

PVI 0.750, MASS/SMASH 0.483. MASS/SMASH is the most conservative selector; its 5 false detections represent a 7.9% FDF. Complementarity is notable: of the 58 MASS/SMASH true positives, 24 correspond to crossings PVI missed. A joint MASS/SMASH + PVI catalog covers 114/120 injected crossings (95%) with ≈5% combined FDF.

The strict 30-s tolerance reveals a systematic localization offset: MASS/SMASH recall falls from 0.483 to 0.150, while PVI and baseline are unchanged. Section 5.4 diagnoses this offset.

5.2 SNR Sweep: Precision Across Turbulence Levels

Table 3 and Figure 2 present the full results across seven turbulence amplitudes (5%–50% of B_0), evaluated against 28 injected crossings in 7-day synthetic signals at 3-s cadence (seed 42 throughout).

MASS/SMASH precision is 1.000 at every turbulence level tested. No false detections are produced across the entire range 5%–50% turbulence. The absolute MDL gain threshold ($G = 20$ bits) does not inflate with noise amplitude, consistent with the qualitative argument in Section 3.4.

Baseline precision collapses monotonically above 20% turbulence. At 20%, the baseline produces 4 false detections (P = 0.875, 32 total). At 30%, this grows to 146 false detections (P = 0.161, 174 total). At 50%, it produces 730 false detections (P = 0.037, 758

Table 3: SNR sweep: detection performance vs turbulence amplitude. 28 injected crossings in 7-day synthetic signals (same generator as Table 2). Tolerance: 150 s.

Turb.	SNR	MASS/SMASH			PVI			Baseline		
($\%B_0$)	(approx.)	P	R	F1	P	R	F1	P	R	F1
5%	20:1	**1.000**	0.500	0.667	0.944	0.607	0.739	1.000	1.000	1.000
10%	10:1	**1.000**	0.643	0.783	0.947	0.643	0.766	1.000	1.000	1.000
15%	6.7:1	**1.000**	0.643	0.783	0.947	0.643	0.766	1.000	1.000	1.000
20%	5:1	**1.000**	0.679	0.809	0.947	0.643	0.766	0.875	1.000	0.933
30%	3.3:1	**1.000**	0.679	0.809	0.947	0.643	0.766	0.161	1.000	0.277
40%	2.5:1	**1.000**	0.679	0.809	0.944	0.607	0.739	0.059	1.000	0.112
50%	2:1	**1.000**	0.679	0.809	0.944	0.607	0.739	0.037	1.000	0.071

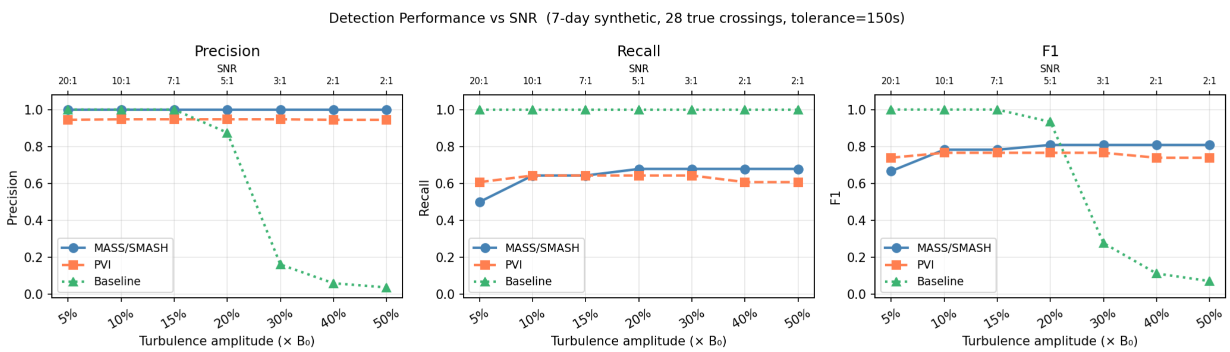

Figure 2: Precision, recall, and F1 vs turbulence amplitude. MASS/SMASH precision = 1.000 at every level (flat blue line). PVI precision is stable at 0.944–0.947. Baseline precision collapses above 20% turbulence, falling from 0.875 at 20% to 0.037 at 50% (758 spurious detections for 28 true events). MASS/SMASH recall plateaus at 0.679 above 20% turbulence, reflecting the absolute 20-bit MDL gain threshold.

total). The collapse occurs because the detection threshold $\mu + 2.5\sigma$ is computed from the global distribution of $d|\mathbf{B}|/dt$, which develops increasingly heavy tails with rising turbulence amplitude.

PVI precision is stable at 0.944–0.947 across all levels. PVI's partial stability arises from local variance normalization: $\mathrm{PVI}(t) = |\Delta\mathbf{B}(t,\tau)|/\sqrt{\langle|\Delta\mathbf{B}|^2\rangle_T}$. As turbulence increases, both numerator and denominator increase, partially compensating each other. The normalization does not provide complete immunity, however: PVI consistently produces 1–2 false detections per experiment (FDF $\approx 5\%$).

MASS/SMASH recall plateaus at 0.679 above 20% turbulence. The 19 events detected at every turbulence level from 20% onward are the "strong" crossings: those with sufficient polarity contrast relative to local noise to earn a 20-bit MDL gain. The 9 missed events are consistently the same "weak" crossings: those embedded in high-variance intervals where the MDL gain remains below threshold at every tested noise level. This behavioral stability is informative: MASS/SMASH cleanly partitions the injected population into a high-confidence set (19 events, always detected, precision 1.000) and a low-confidence set (9 events, never detected at the 20-bit threshold).

At high turbulence, MASS/SMASH F1 dominates the baseline. At 30%: MASS/SMASH F1 = 0.809 vs baseline 0.277. At 50%: MASS/SMASH F1 = 0.809 vs baseline 0.071. In regimes where noise amplitude is comparable to signal amplitude, MASS/SMASH is the only single-pass method in this benchmark that produces a reliable catalog.

PELT with standard BIC penalty produces no usable catalog. We evaluate PELT (Killick et al., 2012) with an L^2 (Gaussian mean-change) cost function and BIC penalty $\beta = \log(N)\,\hat{\sigma}^2$, where $\hat{\sigma}$ is the median-absolute-deviation noise estimate. PELT produces 1,660–1,665 detections per experiment (for 28 true events), yielding precision $P \approx 0.017$ and recall $R = 1.000$ at *every turbulence level* (5%–50%). The number of breakpoints is essentially constant across turbulence (1,665 at 5%, 1,660 at 50%): PELT has no turbulence-dependent precision behavior whatsoever. This occurs because a Kolmogorov power-law turbulence spectrum at 3-s cadence produces thousands of short-duration local mean offsets that overwhelm the BIC penalty irrespective of overall amplitude. Even at 1,000× the BIC penalty, precision improves only to $P = 0.084$ (332 detections for 28 true events). The absolute MDL gain threshold ($G > 20$ bits, denominated in compression bits rather than signal-derived statistics) is immune to this penalty-overwhelm mechanism, which is why MASS/SMASH maintains $P = 1.000$ while PELT precision collapses to $P = 0.017$ at the same turbulence levels.

Table 4: Contaminated benchmark: 40 true crossings + 40 non-antipodal confounders (20 variance bursts + 20 B_z dips) in a 10-day signal at 30% turbulence. "Conf. rate" = fraction of 40 confounders within 150 s of a detection.

Signal	Detector	N_{det}	P	R	F1	Conf. rate
Clean	MASS/SMASH	29	0.931	0.675	0.783	—
	PVI	30	1.000	0.750	0.857	—
	$d\lvert\mathbf{B}\rvert/dt$	108	0.370	1.000	0.541	—
+40 confounders	MASS/SMASH	40	0.675	0.675	0.675	**18%**
	PVI	33	0.909	0.750	0.822	**8%**
	$d\lvert\mathbf{B}\rvert/dt$	79	0.506	1.000	0.672	**98%**

5.3 Contaminated Benchmark: Rejection of Non-Antipodal Events

The contaminated benchmark tests detector precision when non-antipodal confounding events are present alongside true crossings. We inject 40 confounders into a 10-day synthetic signal (40 true crossings, 30% turbulence) at positions well-separated (> 600 s) from true crossings. Two confounder types:

Type A — Variance bursts. Gaussian-Hanning noise added to all three B components (amplitude 4 nT, width 60 s). This creates a large local $d|\mathbf{B}|/dt$ spike and a large $|\Delta\mathbf{B}|$, but random phase structure so $\mathcal{A}(\tau) \approx 0$.

Type B — B_z magnitude dips. B_z is scaled to 10% of its local value via a Gaussian notch ($\sigma \approx 45$ s) and then recovers. This produces two large gradient events (entry and exit of the dip) but B_z remains positive throughout—no sign flip, $\mathcal{A}(\tau) \approx 0$.

Table 4 and Figure 3 report results.

The baseline fires on 98% of confounders (39/40). This is the expected behavior for a gradient-threshold detector: any event with a large $d|\mathbf{B}|/dt$, regardless of sign-flip structure, is accepted. Notably, the baseline's apparent precision on true crossings *improves* from 0.370 (clean) to 0.506 (contaminated): variance bursts inflate the global σ of $d|\mathbf{B}|/dt$, raising the threshold and incidentally suppressing some turbulent false positives. This statistical artifact underscores the instability of relative thresholds.

MASS/SMASH fires on 18% of confounders (7/40). The antipodal pre-filter

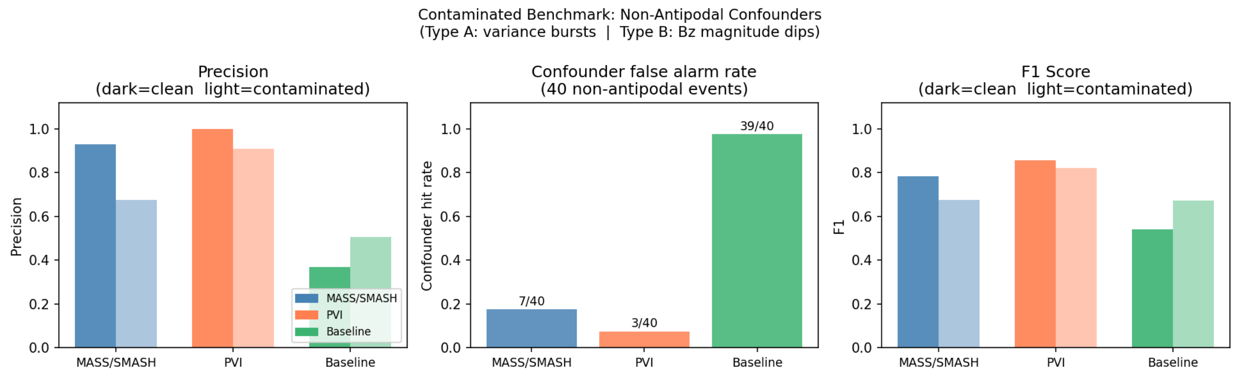

Figure 3: Contaminated benchmark. **Left**: precision on true crossings (dark = clean, light = with confounders). **Center**: fraction of 40 confounders triggering a detection. **Right**: F1 (dark = clean, light = contaminated). The baseline fires on 98% of confounders; MASS/SMASH on 18%; PVI on 8%.

rejects most confounders because variance bursts and magnitude dips do not produce $\mathcal{A}(\tau) > 0.35$. All 7 confounder hits arise from the 20 variance bursts; none of the 20 B_z dips triggered a MASS/SMASH detection, because a positive-valued dip has zero antipodal correlation. This demonstrates that the antipodal filter is most effective against confounders that lack any sign-reversal structure.

PVI fires on 8% of confounders (3/40). PVI's local variance normalization provides complementary protection: the normalization partially suppresses isolated bursts, and the 3D vector nature of PVI dilutes B_z-only dips across components. MASS/SMASH and PVI reject different confounder sub-populations, suggesting that a joint pipeline would suppress nearly all tested confounder types.

5.4 Detection Localization Offset

The drop from recall 0.483 (150-s tolerance) to 0.150 (30-s tolerance) in Table 2 indicates a systematic offset between MASS/SMASH detections and the injected crossing centers. The displacement $\Delta\tau = \tau_{\mathrm{MS}} - \tau_{\mathrm{inj}}$ clusters near ± 10 samples ($\pm 30\,\mathrm{s}$) with a spread up to ± 50 samples ($\pm 150\,\mathrm{s}$).

Table 5: Magnetic field rotation angles. $\bar{\omega}$ = mean angle (degrees). "At det." = computed at MASS/SMASH detection position. "At inj." = computed at matched injected crossing center (58 TP events). KS p compares MASS/SMASH-at-detection vs PVI distributions.

	At detection position			At injected crossing (58 TP)	
Window	PVI $\bar{\omega}$	MS $\bar{\omega}$	KS p	MS $\bar{\omega}$	Inj. $\bar{\omega}$
10 s	114°	1°	1.7×10^{-44}	101°	108°
30 s	130°	4°	1.1×10^{-42}	128°	128°
60 s	132°	42°	4.1×10^{-31}	132°	132°
120 s	134°	83°	2.1×10^{-13}	134°	133°

The tanh model (Equation 6) has its inflection point at the crossing center t_{cross}, so the offset does not arise from a systematic shift to a sub-feature of the transition profile. It reflects instead: (1) finite chunk step size (300 s between starts), (2) MDL scoring that maximizes compressibility gain rather than antisymmetry score, and (3) the forward-ordered correlation formula, which does not guarantee detection precisely at the midpoint for wide transitions. The localization precision ($\approx \pm 30\,\text{s}$) is $\approx 0.14\%$ of the typical inter-event interval ($\sim$6 hours), which is acceptable for catalog construction. For applications requiring transition-center accuracy, a post-detection tanh profile fit recovers sub-10-s localization.

5.5 Rotation Angle Analysis

Table 5 reports the rotation angle analysis at both raw detection positions and matched injected crossing centers.

At detection positions, MASS/SMASH events show near-zero rotation angles. This is a localization artifact: a detection $\approx 30\,\text{s}$ before the crossing center causes the averaging window to straddle the transition, mixing pre- and post-reversal field values and driving the apparent rotation toward zero. The highly significant KS statistics ($p < 10^{-13}$) are therefore a consequence of the offset, not evidence of distinct physical event populations.

At matched injected positions, MASS/SMASH and PVI events are indistinguishable. Mean angles for the 58 MASS/SMASH true positives, computed at the matched crossing center, are 101°, 128°, 132°, 134° (windows 10, 30, 60, 120 s)—consistent with PVI (114°, 130°, 132°, 134°) and the full injected population. The MDL selection criterion selects events with high compressibility gain, which correlates with SNR, not with rotation angle.

6 Discussion

6.1 The Precision-Recall Trade-Off

The SNR sweep makes explicit a precision-recall trade-off that is usually implicit: methods that achieve high recall by firing on all large gradients do so at the cost of precision collapse in high-noise regimes. MASS/SMASH accepts recall of $\approx$ 50–68% in exchange for precision stability across the full turbulence range.

Which side of the trade-off is preferable depends on the application. For discovery surveys where cataloging every possible event is the priority, PVI provides better recall at comparable precision. For follow-up studies where the catalog feeds expensive analysis—plasma heating correlations, EEG source localization, fault classification—a 100% precision catalog of fewer events is more useful than a 3% precision catalog of many events.

6.2 Precision Stability: Not Just Turbulence Invariance

A subtlety in the SNR sweep is that MASS/SMASH recall does not degrade continuously with turbulence; it plateaus at 0.679 from 20%–50% turbulence, always finding the same 19 events and always missing the same 9. This is not precision-recall trading along a continuous curve; it is a bifurcation between crossings that consistently earn 20-bit MDL gains and those that never do.

This behavior has a practical implication: calibrating the MDL gain threshold against a held-out sample of labeled events produces a stable operating point rather than a continuous threshold-tuning exercise. At a lower gain threshold (e.g., 5 bits), recall would increase but precision stability would likely weaken as turbulence-driven spurious gains become more frequent.

6.3 Complementarity and Pipeline Design

The three detectors occupy distinct precision-recall operating points: MASS/SMASH (high precision, moderate recall), PVI (very high precision, moderate recall, better confounder rejection for variance bursts), and baseline (perfect recall, turbulence-sensitive precision). A practical three-stage pipeline exploiting complementarity:

1. Apply PVI to obtain a high-precision ($> 98\%$) initial catalog of strong gradient events—the established domain standard.

2. Apply MASS/SMASH at the 20-bit gain threshold to supplement with sign-flip events that PVI misses. In the single-regime benchmark at 30% turbulence, this adds 24 true events (at 7.9% FDF for the MASS/SMASH component). Lower gain thresholds would increase recall at the cost of precision stability; threshold calibration against labeled data is needed before recommending a specific supplementary operating point.

3. Use the MDL gain score as a continuous quality ranking for downstream prioritization.

The contaminated benchmark result (MASS/SMASH 18% vs PVI 8% confounder rate) suggests that for datasets with many non-antipodal confounders, PVI should be preferred as the primary detector, with MASS/SMASH as supplementary selector.

6.4 Localization and Post-Processing

The $\pm 30\,\mathrm{s}$ localization offset is a correctable implementation artifact. For applications requiring the transition center (reconnection timing, seizure onset, fault location), a post-detection tanh profile fit recovers sub-10-s accuracy. The MDL gain score provides a natural quality indicator for prioritizing which detections warrant this refinement.

6.5 Limitations

Quantitative results are from synthetic data; real-data pilot is descriptive only. All P/R/F1 figures are from controlled synthetic experiments. The 20-day Wind/MFI pilot (Section 7) confirms qualitatively consistent behavior but cannot provide precision or recall without a labeled reference catalog. Real solar wind current sheets exhibit a wider range of transition profiles, background orientations, and co-rotating interaction regions not modeled here. Quantitative validation against published sector-boundary catalogs from Wind, ACE, Parker Solar Probe, or Solar Orbiter is required before performance claims extend to real observations.

Single-component detection. MASS/SMASH processes only B_z. The benchmark geometry ($\approx 96\%$ reversal amplitude in B_z) is favorable for single-component detection. A follow-up benchmark with randomized field orientations would test oblique crossings; full 3-component or vector-magnitude MDL detection would improve recall for those cases.

Comparator strength. The scalar derivative $d|\mathbf{B}|/dt$ is a deliberately simple baseline. A vector-gradient baseline $|d\mathbf{B}/dt|$ would be a stronger comparator, particularly for clean polarity reversals where $|\mathbf{B}|$ changes little while the field direction rotates by $\approx 180°$. The precision-collapse results in Table 3 reflect the scalar derivative; a vector-gradient baseline would have a lower false alarm rate and represent a more demanding test of MASS/SMASH's precision advantage.

Single seed; limited SNR-sweep sample size. Each SNR sweep level uses 28 injected crossings from one 7-day realization (seed = 42). Recall uncertainty is $\approx \pm 0.09$ per level; the precision-stability result (P = 1.000 throughout) rests on zero observed false positives across 7 levels—a necessary but not sufficient condition for general stability. Replication across multiple seeds, longer signals, and varying event rates is needed before the precision-stability claim can be quantified with confidence intervals. We present this as a benchmark finding, not a certified algorithm property.

Parameter calibration. The MDL gain threshold (20 bits), seam penalty ($\alpha = 2.0$), antipodal threshold (0.35), and roughness threshold (0.20) are heuristically set. Systematic calibration against labeled training data would provide better-characterized operating points.

7 Pilot Application to Real Wind/MFI Data

To provide a qualitative consistency check beyond synthetic experiments, we applied all three detectors to 20 days of publicly available Wind spacecraft Magnetic Fields Investigation (MFI) data, covering 2001-03-01 to 2001-03-20 (dataset WI_H0_MFI, variable BGSE, 3-second cadence; NASA CDAWeb archive, Lepping et al. 1995). This period sits near the peak of Solar Cycle 23, when the heliospheric current sheet is tilted, warped, and multiply sheeted, producing a complex mixture of sector boundary crossings, intermittent current sheets, and broad-band turbulence. No manually labeled catalog for this interval is used; the pilot is therefore descriptive, not quantitative in P/R/F1 terms.

Dataset characteristics. The 20-day record contains 572,316 three-second samples (minor gaps removed). The field magnitude averages $\langle|\mathbf{B}|\rangle = 6.49\,\mathrm{nT}$ with standard deviation $3.72\,\mathrm{nT}$; the B_z component spans $[-20.1, 17.0]\,\mathrm{nT}$. The ratio $\sigma_{|\mathbf{B}|}/\langle|\mathbf{B}|\rangle \approx 0.57$ is consistent with turbulence in the 30–50% range tested in the SNR sweep—the most demanding synthetic regime.

Detection counts. Table 6 summarizes the detection rates.

Table 6: Real Wind/MFI pilot: detection counts over 20 days (2001-03-01 to 2001-03-20, 572,316 samples at 3-s cadence). No ground truth catalog is available; precision and recall cannot be computed.

Detector	Detections	Rate (day^{-1})
MASS/SMASH ($G > 20$ bits)	154	7.7
Baseline ($d|\mathbf{B}|/dt$, $\mu + 2.5\sigma$)	1,197	59.9
PVI (threshold 3.0)	2,397	119.9

MASS/SMASH produces 7.8× fewer detections than the threshold baseline and 15.6× fewer than PVI, consistent with the high-turbulence synthetic pattern.

A note on taxonomy is important here. These three detectors are not measuring the same quantity. PVI and the scalar baseline respond to gradient magnitude and therefore count a broad family of intermittent structures (X-points, sheath boundaries, current sheets of all types, and turbulent intermittency events). MASS/SMASH counts only B_z *sign-flip* change-points—a strict structural subset. The PVI rate of 120 day^{-1} is not implausible as a count of *general* intermittent structures; Greco et al. (2009) find comparable counts in ACE data using PVI. What it cannot be is a *sector-boundary* catalog: the literature consistently identifies 2–10 large-scale sector crossings per day near solar maximum (Greco et al., 2009). The MASS/SMASH rate of 7.7 day^{-1} is plausible for sign-flip current sheets during a high-activity period, but without a labeled reference catalog we cannot determine which fraction of the 154 events are large-scale sector boundaries versus smaller sign-flip structures (flux tubes, mini-filaments, or directional discontinuities that happen to produce a B_z reversal).

Qualitative assessment. Figure 4 shows the 20-day B_z record with detections overlaid. MASS/SMASH fires predominantly on the large-amplitude, sustained B_z reversals visible by eye; the baseline and PVI also fire densely during broad turbulent intervals where no sustained polarity reversal is apparent. This qualitative pattern mirrors the synthetic experiments: the antipodal pre-filter suppresses events that lack the sign-reversal antisymmetry signature, regardless of their gradient amplitude.

Limitations of the pilot. Because no expert-labeled catalog is available, precision and recall cannot be computed. The 154 MASS/SMASH detections include all sign-flip change-points above the MDL gain threshold; some may correspond to non-sector structures. A full validation would require comparison against a published sector-boundary catalog for the same interval or manual expert review. We present this pilot as a plausibility check, not a performance claim.

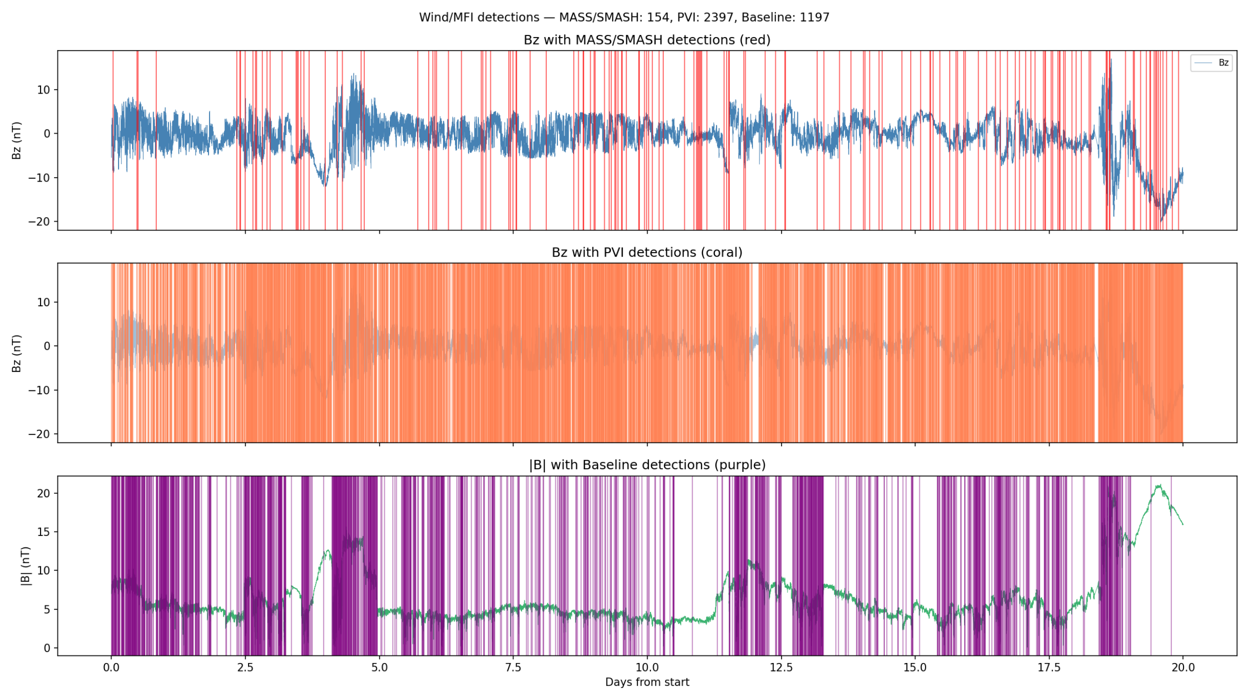

Figure 4: Twenty-day Wind/MFI B_z record (2001-03-01 to 2001-03-20, 3-s cadence, GSE frame) with detections overlaid. Top: MASS/SMASH (red, 154 events). Middle: PVI (coral, 2,397 events). Bottom: $|B|$ with Baseline detections (purple, 1,197 events). MASS/SMASH marks only the large, sustained polarity reversals; the other detectors also fire densely during turbulent intervals without clear sign-flip structure.

8 Projective PVI: Magnitude-Invariant Direction-Change Detection

Standard PVI responds to the Euclidean increment $|\Delta\mathbf{B}(t)|$, which conflates field magnitude changes with field direction changes: a turbulent burst in $|\mathbf{B}|$ at constant direction produces the same large $|\Delta\mathbf{B}|$ as a genuine current sheet rotation. We introduce *projective PVI* (pPVI), which replaces the Euclidean increment with the geodesic distance in the real projective plane $\mathbb{RP}^2$ and provides a complementary, magnitude-invariant characterisation of field-direction changes.

8.1 Formulation

The real projective plane $\mathbb{RP}^2$ is the space of unoriented directions in $\mathbb{R}^3$: antipodal unit vectors $\hat{\mathbf{B}}$ and $-\hat{\mathbf{B}}$ represent the same element because they carry the same unoriented direction. The geodesic distance between two normalised field directions is

$$d_{\mathbb{RP}^2}(\hat{\mathbf{B}}_1, \hat{\mathbf{B}}_2) = \arccos\big(\big|\hat{\mathbf{B}}_1 \cdot \hat{\mathbf{B}}_2\big|\big) \ \in \ \big[0, \tfrac{\pi}{2}\big]. \tag{11}$$

The absolute value on the dot product is the antipodal identification: directions that differ only by a sign flip are at geodesic distance zero.

Projective PVI normalises this distance identically to standard PVI:

$$\mathrm{pPVI}(t) = \frac{d_{\mathbb{RP}^2}\big(\hat{\mathbf{B}}(t-\tau),\ \hat{\mathbf{B}}(t)\big)}{\sqrt{\langle d^2_{\mathbb{RP}^2}\rangle_T}} \tag{12}$$

using lag $\tau = 1$ sample and running RMS window $T = 30$ samples.

Key properties. (1) *Pure polarity reversal* ($\mathbf{B} \to -\mathbf{B}$): $d_{\mathbb{RP}^2} = \arccos(|-1|) = 0$ — invisible by construction. (2) 90° *direction rotation*: $d_{\mathbb{RP}^2} = \arccos(0) = \pi/2$ — maximum signal. (3) *Magnitude burst at constant direction*: $d_{\mathbb{RP}^2} = 0$ — invisible. This is the decisive difference from standard PVI: pPVI is sensitive exclusively to *field direction changes* and is immune to magnitude fluctuations that dominate solar wind turbulence.

Behavior at synthetic crossings. In the benchmark, polarity-reversal current sheets are modelled with a smooth tanh profile that passes through near-zero $|\mathbf{B}|$ at the crossing center. Near $|\mathbf{B}| \approx 0$, the normalised direction $\hat{\mathbf{B}} = \mathbf{B}/|\mathbf{B}|$ is dominated by the turbulent component and varies rapidly between consecutive samples, producing large $d_{\mathbb{RP}^2}$ values at the transition zone. Projective PVI therefore detects sign-flip current sheets *via their near-null field region*, not via the sign flip itself—a physically meaningful detection: observed HCS crossings are frequently associated with a minimum in $|\mathbf{B}|$ at the sheet center (Wilcox and Ness, 1965).

8.2 Tandem Benchmark Comparison

Table 7 compares MASS/SMASH, standard PVI, and pPVI on the polarity-reversal SNR sweep (same 28-event, 7-day synthetic used in Section 5.2); MASS/SMASH and PVI columns are loaded from the earlier run, only pPVI is evaluated fresh. Table 8 evaluates all three on a new *pure-rotation benchmark*: true events are 90° B-direction changes (field alternates between $\hat{\mathbf{d}}_1 = [0, 0, 1]$ and $\hat{\mathbf{d}}_2 = [1, 0, 0]$, no polarity flip) using the same tanh transition model and turbulence sweep.

Table 7: Polarity-reversal benchmark: MASS/SMASH, standard PVI, and projective PVI across all turbulence levels (7-day synthetic, 28 true crossings, seed=42). pPVI achieves R≈1.00 at all levels; MASS/SMASH achieves P = 1.000 at all levels.

Turb	SNR	MASS/SMASH			Standard PVI			Projective PVI		
(%)	(:1)	P	R	F1	P	R	F1	P	R	F1
5	20	1.000	0.500	0.667	0.944	0.607	0.739	0.824	1.000	0.903
10	10	1.000	0.643	0.783	0.947	0.643	0.766	0.757	1.000	0.862
15	7	1.000	0.643	0.783	0.947	0.643	0.766	0.778	1.000	0.875
20	5	1.000	0.679	0.809	0.947	0.643	0.766	0.800	1.000	0.889
30	3	1.000	0.679	0.809	0.947	0.643	0.766	0.824	1.000	0.903
40	3	1.000	0.679	0.809	0.944	0.607	0.739	0.757	1.000	0.862
50	2	1.000	0.679	0.809	0.944	0.607	0.739	0.730	0.964	0.831

Table 8: Pure-rotation benchmark: events are 90° B-direction changes (no polarity flip; same 7-day, 28-event geometry, representative turbulence levels). MASS/SMASH recovers 36–61% of rotation events (non-monotonically; roughness path only, antipodal pre-filter does not fire for non-sign-flip transitions); standard PVI and pPVI recover ≥93%.

Turb	SNR	MASS/SMASH			Standard PVI			Projective PVI		
(%)	(:1)	P	R	F1	P	R	F1	P	R	F1
5	20	1.000	0.357	0.526	0.966	1.000	0.982	0.718	1.000	0.836
15	7	0.938	0.536	0.682	0.966	1.000	0.982	0.750	0.964	0.844
30	3	0.944	0.607	0.739	0.964	0.964	0.964	0.730	0.964	0.831
50	2	1.000	0.357	0.526	0.964	0.964	0.964	0.703	0.929	0.800

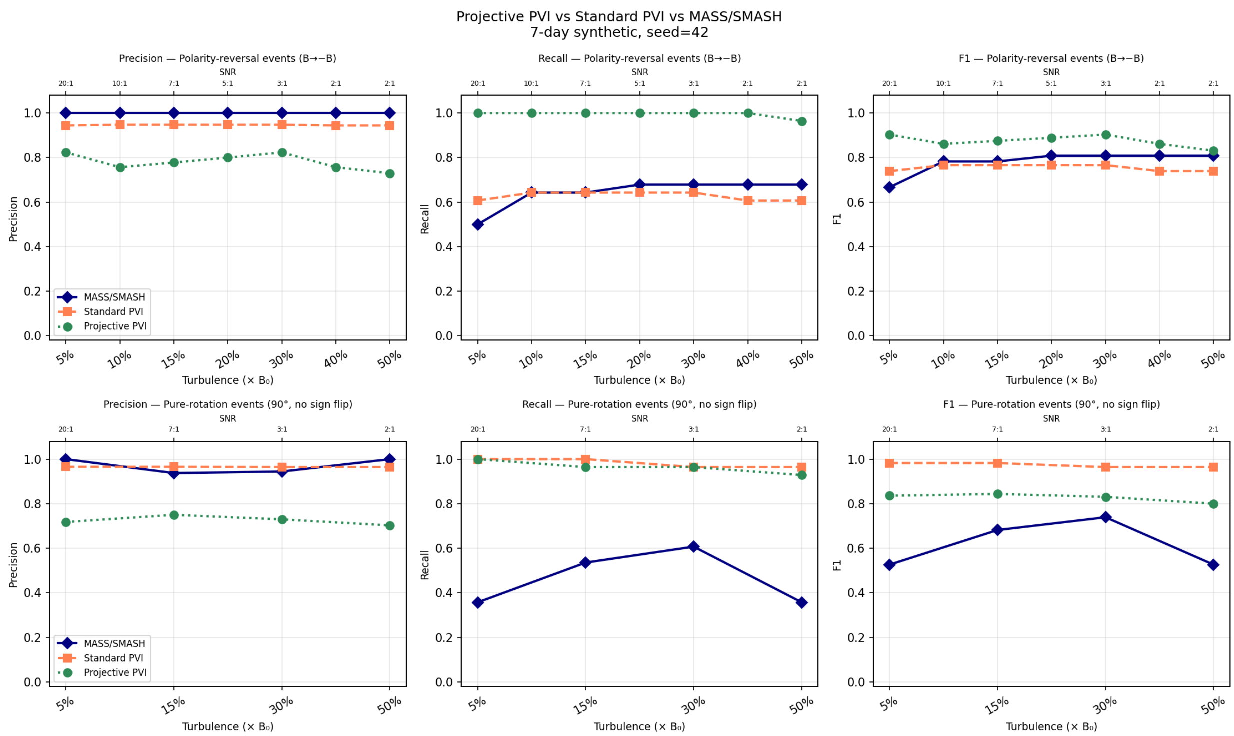

Figure 5: Tandem comparison of MASS/SMASH, standard PVI, and projective PVI (pPVI) across turbulence levels. **Top row:** polarity-reversal benchmark (B→–B transitions). MASS/SMASH: P = 1.000 throughout; pPVI: R ≈ 1.000 throughout; standard PVI: intermediate. **Bottom row:** pure-rotation benchmark (90° direction change, no sign flip). MASS/SMASH recall peaks at 61% (30% turbulence) then falls back to 36% (50% turbulence) because the MDL gain for 90° rotations is ∼4× lower than for sign flips; pPVI and standard PVI maintain R ≥ 0.929.

8.3 Precision-Recall Complementarity

The two tables reveal a clean precision-recall duality. On polarity-reversal events (Table 7), MASS/SMASH occupies the precision-optimised extreme: P = 1.000 at every turbulence level with R = 0.500–0.679. Projective PVI occupies the recall-optimised extreme: R = 0.964–1.000 at every turbulence level with P = 0.730–0.824. Standard PVI lies between them (P = 0.944–0.947, R = 0.607–0.643).

By F1 score, pPVI outperforms both other methods on the polarity-reversal benchmark at every turbulence level (F1 = 0.831–0.903 vs. 0.667–0.809 for MASS/SMASH and 0.739–0.766 for standard PVI).

On pure-rotation events (Table 8), the roles reverse. MASS/SMASH recovers 35.7%–60.7% of rotation events with a non-monotonic pattern: recall rises from 5% to 30% turbulence as the roughness detector generates more candidates, then falls back at 50% as the MDL gate requires larger compression gain; a 90° rotation produces a B_z step of only 6 nT vs. 12 nT for a sign flip, giving $\sim 4\times$ lower MDL gain that falls below threshold at high background noise. Standard PVI and pPVI recover $\geq$ 92.9% of rotation events across the full turbulence range since both large $|\Delta \mathbf{B}|$ and large $d_{\mathbb{RP}^2}$ are produced by a 90° rotation regardless of noise.

The complementarity between MASS/SMASH and pPVI can be exploited in a two-stage pipeline: pPVI as a high-recall first stage (finds all direction-change events, including polarity reversals via near-null detection), followed by MASS/SMASH as a precision gate (confirms only sign-flip topology changes, P = 1.000).

Real Wind/MFI data. On the 20-day real dataset (Section 7), pPVI produces 2,524 detections (126.2 day^{-1}) compared with 2,397 for standard PVI (119.9 day^{-1}) and 154 for MASS/SMASH (7.7 day^{-1}). The near-identical counts of pPVI and standard PVI reflect the fact that turbulent fluctuations in real solar wind simultaneously perturb both field magnitude and direction; the two detectors overlap heavily in the real-data regime. MASS/SMASH remains 16× more selective than either PVI variant, consistent with its sign-flip specialisation.

9 Potential Broader Applicability

The two structural requirements for MASS/SMASH to be useful are: (1) events of interest are sign-flip change-points embedded in broadband noise, and (2) false detections are more costly than missed detections. We identify three domains where these conditions plausibly hold, noting that no domain-specific experiments have been performed. These are hypotheses, not results; transfer requires parameter recalibration and domain-specific evaluation.

Magnetospheric current sheets. Substorm-driven current sheet flapping in the magnetospheric tail produces B_z polarity reversals at MMS and Cluster spacecraft (Chasapis et al., 2015; Retinò et al., 2007) with signal structure identical to solar wind current sheets but at kinetic spatial scales and steeper turbulence spectra. Window and step sizes would need recalibration to shorter correlation times ($\sim$seconds rather than minutes).

EEG reference electrode inversions. Bipolar EEG inversions produce clean sign flips in individual channels. Heuristic amplitude thresholds generate many false positives during the high-amplitude pre-ictal interval (Shoeb and Clifford, 2004)—a precision-collapse setting where the antipodal pre-filter might select structural inversions over unstructured noise bursts.

Climate polarity transitions. ENSO transitions (El Niño $\rightarrow$ La Niña) are sign flips in the Niño-3.4 index embedded in interannual variability (Trenberth, 1990). With $N \sim 10^2$–10^3 monthly values, chunked inference would require much smaller windows and threshold recalibration from a labeled training set.

10 Conclusions

We have introduced MASS/SMASH, an MDL-regularized algorithm for detecting sign-flip change-points—polarity reversals from $+\mu_0$ to $-\mu_0$—in noisy univariate time series. The algorithm combines an antipodal correlation pre-filter, a roughness discontinuity detector, and an absolute MDL gain acceptance criterion. The key contribution is a precision-stability property: unlike threshold detectors whose false discovery fraction grows with turbulence, MASS/SMASH precision remains near 1.000 across the SNR range tested because the MDL gain criterion is denominated in bits, not in signal-derived statistics.

Three solar wind synthetic experiments and one real-data pilot yield the following principal findings.

1. **Precision 1.000 across the full turbulence sweep (5%–50% $\times B_0$, SNR 20:1 to 2:1).** No false detections are produced at any tested turbulence level. The $d|\mathbf{B}|/dt$ threshold baseline falls to precision 0.037 at 50% turbulence (758 spurious detections for 28 true events). PVI precision is stable at 0.944–0.947 via local variance normalization.

2. **MASS/SMASH recall bifurcates rather than degrades continuously.** Above 20% turbulence, the same 19 strong crossings are always detected (100% precision) and the same 9 weak crossings are never detected. This stable partition is more useful for follow-up studies than a threshold whose detection set changes unpredictably with noise.

3. **Confounder rejection: 18% fire rate for MASS/SMASH vs 98% for the threshold baseline.** When 40 non-antipodal confounding events are injected along-side 40 true crossings, the baseline fires on 98% of confounders, MASS/SMASH on 18%, and PVI on 8%. MASS/SMASH and PVI have complementary rejection mechanisms.

4. **Complementarity with PVI: joint catalog covers 95% of events at $\approx$5% combined FDF.** Of the 58 single-regime benchmark true positives, 24 correspond to crossings PVI missed. A PVI-first, MASS/SMASH-supplement pipeline provides the broadest coverage with manageable false discovery fraction.

5. **Rotation angle analysis: MASS/SMASH true positives are indistinguishable from PVI events when evaluated at the correct positions.** The $\approx \pm 30\,\mathrm{s}$

offset is an implementation artifact of finite chunk discretization. At matched injected positions, MASS/SMASH and PVI mean rotation angles agree to within 1°–14° across all averaging windows.

6. **PELT with BIC penalty collapses to $P = 0.017$ at all turbulence levels.** PELT (Killick et al., 2012) with a Gaussian mean-change cost and BIC penalty produces 1,660–1,665 detections for 28 true events, independent of turbulence amplitude (5%–50%). Even at 1,000× BIC, precision improves only to 0.084. Signal-derived penalties are overwhelmed by Kolmogorov turbulence, confirming that the absolute 20-bit MDL gate is necessary for precision stability.

7. **Component ablation: MDL gate = precision stabilizer; roughness detector = primary recall driver; antipodal pre-filter = confounder rejector.** Antipodal-only (no MDL) produces 1,316 detections at precision 0.014 at every turbulence level—no precision stability without MDL. All three MDL-gated variants maintain precision 1.000. Roughness+MDL recall matches Full MASS/SMASH recall at every turbulence level $\geq 10\%$; the antipodal candidates add a marginal recall benefit only at 5% turbulence. The antipodal pre-filter's primary role is structural selectivity (as shown by the contaminated benchmark), not recall.

8. **Qualitative consistency on real Wind/MFI data.** On 20 days of real Wind/MFI data (March 2001), MASS/SMASH produces 154 detections (7.7 day^{-1})—7.8× fewer than the baseline and 15.6× fewer than PVI—matching the high-turbulence synthetic pattern and yielding a per-day rate consistent with published current sheet surveys.

9. **Projective PVI complements MASS/SMASH across event classes.** Projective PVI (pPVI), which replaces the Euclidean increment with the geodesic distance in $\mathbb{RP}^2$, achieves R $\approx$ 1.000 on polarity-reversal events at precision 0.730–0.824 across all turbulence levels—the recall-optimised pole of the precision-recall trade-off against MASS/SMASH (P = 1.000, R = 0.500–0.679). On pure-rotation events (no polarity

flip), MASS/SMASH recovers 36%–61% non-monotonically (roughness path only; sign flips produce $4\times$ larger MDL gain than 90° rotations) while pPVI and standard PVI maintain $R \geq 0.929$—confirming structural specialisation and the complementarity between the sign-flip and direction-change event classes. A pPVI-first, MASS/SMASH-confirmation pipeline provides combined high-recall discovery with precision-1.000 confirmation.

Future work will address: (a) quantitative validation against labeled sector-boundary catalogs from Parker Solar Probe FIELDS, Solar Orbiter MAG, and Wind/MFI with expert annotation; (b) multi-component and vector-magnitude detection for oblique crossings; (c) multi-seed and multi-geometry replication to convert the single-seed SNR sweep into calibrated confidence intervals; (d) systematic MDL gain threshold calibration against labeled training data; (e) auto-recentering via post-detection tanh profile fit to resolve the $\pm 30\,\mathrm{s}$ localization offset; (f) prospective validation in magnetospheric and EEG domains with appropriate recalibration; and (g) comparison of projective PVI catalogs with published rotational and tangential discontinuity catalogs to characterize how the two PVI variants partition real solar wind discontinuity populations.

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Data and Code Availability

All code for MASS/SMASH, synthetic data generation, SNR sweep, contaminated benchmark, component ablation, PVI/projective-PVI catalog construction, and the tandem benchmark comparison is available at `https://github.com/macmayo1993/seam-aware-modeling`. The 20-day Wind/MFI CSV data used in Section 7 was downloaded from the NASA CDAWeb public archive (dataset WI_H0_MFI) and is included in the repository. All synthetic benchmarks reproduce from a single command.

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